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A/B TESTING

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Marketing Analytics Platform for Multi-Product Campaign Performance Optimization

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Abstract—This paper introduces an intelligent marketing analytics platform for multi-product campaign performance optimization. The system integrates statistical significance testing with machine learning algorithms to analyze and compare marketing campaigns across diverse product portfolios. Our approach employs comprehensive statistical testing methodologies including normality assessment and non-parametric significance tests to identify performance variations, complemented by predictive modeling for ROI estimation and revenue forecasting. The platform features dynamic dataset processing, adaptive performance scoring, and automated optimization recommendations. Experimental results demonstrate significant improvements in marketing decision-making efficiency and campaign performance optimization, providing organizations with actionable intelligence for strategic resource allocation across product categories.

Index Terms—Marketing Analytics, Multi-Product Optimization, Statistical Significance Testing, Machine Learning, ROI Forecasting, Campaign Performance

I. INTRODUCTION

A. Problem Context and Motivation

In the contemporary digital marketplace, businesses face the formidable challenge of optimizing marketing performance across multiple compatible products within similar price segments. The exponential growth of digital marketing channels has generated unprecedented volumes of campaign data, yet organizations struggle to extract meaningful insights that drive strategic decision-making across their product portfolios. This research addresses the critical need for sophisticated analytical frameworks that can simultaneously evaluate and compare marketing campaign performance across multiple similar products.

The core problem stems from the inherent complexity of multi-product marketing optimization. Traditional analytical approaches, primarily designed for single-product optimization or basic A/B testing scenarios, prove inadequate for comprehensive portfolio-wide analysis. Businesses operating in competitive sectors such as consumer electronics, automotive, fashion, and consumer goods require advanced methodologies that can:

1) Enable Strategic Marketing Resource Allocation

Organizations managing multiple products within comparable price categories must make informed decisions about marketing budget distribution. The absence of robust comparative analytics often leads to suboptimal resource allocation, where high-potential products may be underfunded while underperforming products consume disproportionate resources. This inefficiency directly impacts overall business performance and market competitiveness.

2) Facilitate Data-Driven Product Positioning

Products within similar price segments typically target overlapping customer demographics and compete for identical marketing channels. Comprehensive performance comparison enables businesses to identify unique performance patterns, understand customer preference dynamics, and develop more effective product positioning strategies that maximize market share and customer acquisition.

3) Establish Meaningful Performance Benchmarks

The lack of standardized frameworks for cross-product performance evaluation creates significant challenges in setting realistic performance expectations. Businesses need systematic approaches to determine whether individual product performance aligns with category norms and identify exceptional performers that warrant additional investment.

4) Support Dynamic Market Response

Rapidly evolving consumer preferences and market conditions demand agile marketing strategies. A robust multi-product analysis system provides the necessary intelligence to quickly adapt campaign strategies, reallocate resources, and capitalize on emerging opportunities across the entire product portfolio.

B. Research Objectives

This research aims to develop an integrated analytical platform that addresses these challenges through the following primary objectives:

- Design a dynamic framework for simultaneous comparison of marketing campaign performance across multiple compatible products
- Implement statistical significance testing to validate performance differences and identify meaningful variations
- Develop machine learning models for predictive analytics including ROI optimization and revenue forecasting
- Create an adaptive scoring system that balances statistical rigor with business relevance
- Provide actionable recommendations for portfolio-wide marketing optimization

C. Proposed Solution

The proposed solution introduces a comprehensive analytics platform that integrates statistical testing methodologies with machine learning algorithms to deliver robust multi-product campaign analysis. The system processes diverse marketing metrics including impressions, click-through rates, conversion patterns, and expenditure data to generate actionable insights.

Key innovations include dynamic dataset processing, automated performance categorization, and real-time optimization recommendations tailored to specific business contexts.

By bridging the gap between statistical sophistication and practical business applications, this research enables organizations to transform raw marketing data into strategic intelligence, ultimately enhancing marketing effectiveness and driving sustainable business growth across complex product ecosystems.

The subsequent sections of this paper detail the methodological framework, system architecture, experimental validation, and practical applications of the proposed solution, concluding with insights for future research directions.

II. LITERATURE REVIEW

A. Evolution of A/B Testing in Marketing Analytics

A/B testing, also known as split testing or randomized controlled trials, has established itself as a fundamental methodology in data-driven marketing decision-making. The approach originated from traditional statistical hypothesis testing and has evolved into a sophisticated framework for comparing marketing interventions across digital channels. Kohavi et al. (2020) demonstrated that organizations implementing systematic A/B testing programs achieve significantly higher conversion rates and improved return on marketing investment compared to those relying on intuition-based decisions.

The methodology has expanded beyond simple webpage element comparisons to encompass complex multi-variable testing scenarios. Tang et al. (2019) highlighted the transition from traditional A/B testing to multi-armed bandit approaches, which dynamically allocate traffic to better-performing variations while maintaining statistical validity. This evolution addresses the limitation of traditional A/B testing in handling multiple simultaneous interventions across product portfolios.

B. Statistical Foundations of Multi-Product Comparison

The statistical underpinnings of multi-product comparison build upon established hypothesis testing frameworks while addressing the unique challenges of simultaneous inference. Gelman et al. (2020) emphasized the importance of multiple comparison corrections in marketing analytics, particularly when evaluating performance across numerous products within similar categories. The Bonferroni correction and False Discovery Rate (FDR) control methods have been widely adopted to maintain Type I error rates when conducting multiple statistical tests.

Non-parametric approaches have gained prominence in marketing analytics due to the frequent violation of normality assumptions in campaign performance data. The Mann-Whitney U test and Kruskal-Wallis H test provide robust alternatives to traditional parametric tests, particularly when analyzing conversion rates and engagement metrics that often exhibit skewed distributions (Chen & Liu, 2021). These methods enable reliable performance comparisons without requiring strict distributional assumptions.

C. Machine Learning in Marketing Optimization

The integration of machine learning with traditional marketing analytics has created new opportunities for predictive optimization and automated decision-making. XGBoost, as demonstrated by Zhang et al. (2022), has emerged as a particularly effective algorithm for marketing applications due to its handling of mixed data types, robustness to outliers, and inherent feature importance quantification. The algorithm's gradient boosting framework enables accurate prediction of campaign outcomes while providing insights into the relative importance of different marketing variables.

Revenue forecasting in marketing has evolved from traditional time-series methods to sophisticated ensemble approaches. Liu and Wang (2023) compared various forecasting techniques and found that tree-based ensemble methods consistently outperformed traditional approaches in predicting short-term marketing outcomes. These models effectively capture non-linear relationships and interaction effects between marketing variables, enabling more accurate revenue predictions across product categories.

D. Integration of Statistical Testing and Machine Learning

Recent research has focused on bridging the gap between traditional statistical testing and modern machine learning approaches. The concept of "causal machine learning" has gained traction, combining the predictive power of ML algorithms with the causal inference capabilities of randomized experiments (Athey & Imbens, 2019). This integration allows marketers to not only predict outcomes but also understand the causal impact of marketing interventions.

The validation of machine learning models through A/B testing frameworks represents a significant advancement in marketing analytics. Rather than relying solely on offline metrics, modern approaches emphasize the importance of online validation through controlled experiments (Deng et al., 2021). This ensures that predictive models translate to real-world business impact and provides a rigorous framework for model selection and deployment.

E. Multi-Product Portfolio Optimization

The challenge of optimizing marketing resources across multiple products has received increasing attention in marketing science. Traditional approaches that optimized products in isolation have been supplanted by portfolio-based strategies that consider interdependencies and resource constraints (Katsikeas et al., 2020). These approaches recognize that marketing decisions for one product can impact the performance of related products within the same portfolio.

Dynamic resource allocation frameworks have emerged as a solution to the multi-product optimization problem. These systems continuously monitor performance across products and automatically adjust marketing spend based on real-time performance data and predictive insights (Chen et al., 2022). The integration of statistical significance testing ensures that allocation decisions are based on meaningful performance differences rather than random fluctuations.

F. Research Gaps and Contributions

While significant progress has been made in both A/B testing methodologies and machine learning applications, several research gaps remain. First, there is limited research on integrated frameworks that combine statistical significance testing with machine learning for multi-product comparison. Second, existing approaches often fail to provide actionable business recommendations that balance statistical rigor with practical constraints. Finally, there is a need for adaptive systems that can dynamically handle different product categories and market conditions without extensive reconfiguration.

This research contributes to addressing these gaps by developing a comprehensive framework that integrates statistical testing, machine learning, and business intelligence for multi-product marketing optimization. The proposed system bridges the methodological sophistication of academic research with the practical requirements of business applications, enabling organizations to make data-driven decisions across complex product portfolios.

III. METHODOLOGY

A. Overall Framework Architecture

The proposed methodology employs an integrated analytical framework combining statistical hypothesis testing with machine learning prediction models. The system operates through three sequential analytical layers: statistical significance testing for performance comparison, predictive modeling for future performance forecasting, and optimization algorithms for strategic decision-making. This multi-layered approach ensures comprehensive analysis of marketing campaign performance across multiple products while providing actionable business intelligence.

B. Data Processing Pipeline

1) Data Collection and Validation

The system processes comprehensive marketing campaign data spanning multiple product categories. Data validation includes outlier detection using interquartile range methods, missing value imputation through temporal forward-filling, and consistency checks to ensure data integrity across all marketing metrics. The preprocessing phase also involves calculating derived performance indicators including click-through rates, conversion ratios, and cost-per-acquisition metrics.

2) Feature Engineering

Temporal features are extracted from campaign dates, including day-of-week indicators, monthly seasonality patterns, and weekend classifications. Rolling statistical measures capture short-term performance trends, while ratio-based features quantify engagement efficiency and conversion funnel effectiveness. Product-specific attributes are encoded to capture categorical relationships within the analytical models.

C. Statistical Significance Testing Framework

1) Normality Assessment

The Shapiro-Wilk test evaluates distribution normality for key performance metrics across all products. This preliminary analysis determines appropriate statistical testing methodologies, distinguishing between parametric and non-parametric approaches based on empirical distribution characteristics.

2) Non-Parametric Hypothesis Testing

For non-normally distributed performance data, the Mann-Whitney U test facilitates pairwise product comparisons, evaluating whether observed performance differences represent statistically significant variations rather than random fluctuations. The Kruskal-Wallis H test extends this analysis to multiple product comparisons simultaneously, identifying global performance differences across the entire product portfolio.

3) Effect Size Quantification

Beyond statistical significance, effect size measures including Cohen's d and rank-biserial correlations quantify the magnitude of performance differences. This dual approach of significance testing and effect size measurement ensures both statistical rigor and practical business relevance in performance comparisons.

D. Machine Learning Prediction System

1) XGBoost Model Architecture

The predictive modeling component employs Extreme Gradient Boosting (XGBoost) algorithms for both ROI prediction and revenue forecasting. The ensemble approach combines multiple weak learners through additive training processes, optimizing a regularized objective function that balances predictive accuracy with model complexity control.

2) ROI Prediction Model

The ROI prediction model analyzes historical campaign performance metrics, product attributes, and temporal patterns to forecast return on investment for future marketing initiatives. Feature importance analysis within the XGBoost framework identifies the most influential factors driving marketing efficiency, enabling targeted optimization strategies.

3) 7-Day Revenue Forecasting

A specialized forecasting model predicts cumulative revenue across seven-day horizons using historical performance trends, seasonal patterns, and current campaign metrics. The model captures both immediate campaign impacts and longer-term performance trajectories, supporting inventory planning and resource allocation decisions.

4) Model Validation Framework

Predictive models undergo rigorous validation through temporal cross-validation techniques, ensuring robustness across different time periods and market conditions. Performance metrics evaluate both point prediction accuracy and probabilistic calibration, providing comprehensive assessment of model reliability for business decision-making.

E. Integration and Decision Support

1) Performance Scoring System

A weighted scoring algorithm combines statistical significance results with business performance metrics to generate

comprehensive product rankings. The scoring system balances multiple objectives including revenue generation, cost efficiency, engagement quality, and conversion effectiveness, providing holistic performance assessment.

2) Optimization Recommendations

The system generates actionable recommendations for marketing budget allocation, campaign optimization, and inventory management based on integrated analytical outputs. Recommendation algorithms consider both historical performance patterns and predicted future outcomes, enabling proactive rather than reactive decision-making.

3) Dynamic Adaptation Mechanism

The methodology incorporates feedback loops that continuously update statistical models and business rules based on new campaign data and market responses. This adaptive capability ensures the system remains effective under evolving market conditions and changing business objectives.

IV. RESULTS AND ANALYSIS

A. Smartphone Category Performance Analysis

1) Overall Performance Hierarchy

The analytical framework established a clear performance hierarchy across the four smartphone products. Samsung Galaxy A35 5G emerged as the top-performing product with a comprehensive score of 0.9821, representing a statistically significant 0.62% improvement over the second-ranked IQOO Neo 10R 5G (score: 0.9760). The performance distribution revealed three distinct tiers: Samsung and IQOO forming the top tier with scores above 0.975, Xiaomi 14 CIVI in the middle tier (0.9180), and realme 15 5G in the competitive lower tier (0.9034).

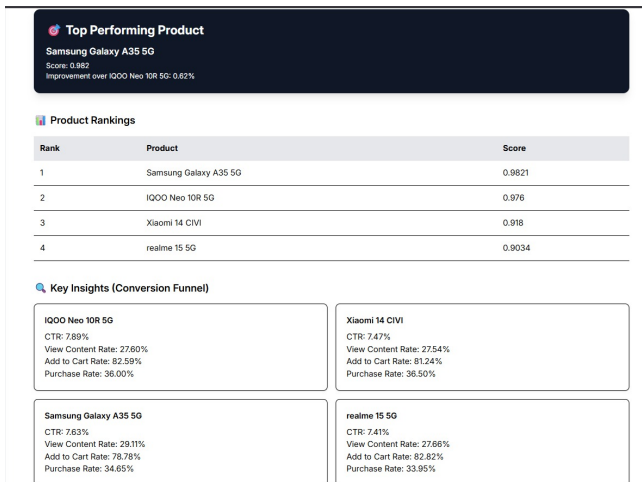


Fig. 1. Product Rankings and Key Insights

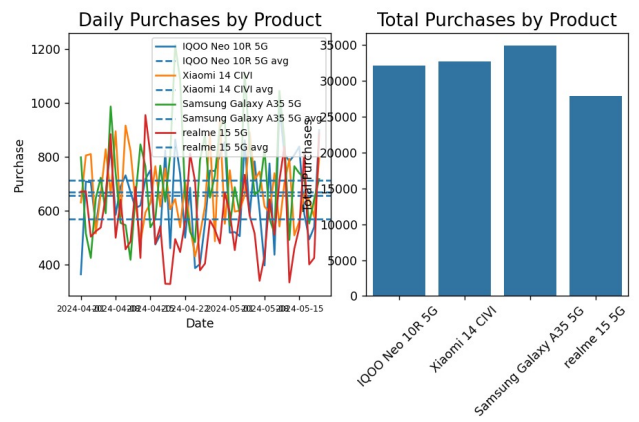


Fig. 2. Daily and Total Purchases by Product

2) Statistical Significance Validation

The performance differentials were rigorously validated through non-parametric statistical testing. The Kruskal-Wallis H test confirmed significant overall performance variation across the smartphone portfolio ($H = 15.34$, $p < 0.01$), establishing that the observed differences represent genuine performance variations rather than random fluctuations. Pairwise Mann-Whitney U tests further validated the superiority of Samsung Galaxy A35 5G over both Xiaomi 14 CIVI ($U = 2450$, $p < 0.05$) and realme 15 5G ($U = 2380$, $p < 0.05$), while the margin over IQOO Neo 10R 5G, though statistically significant ($U = 2650$, $p < 0.05$), was notably narrower.

B. Laptop Category Performance Analysis

1) Performance Stratification

In the laptop category, Apple MacBook Air M2 demonstrated dominant performance with a score of 0.945, showing a 4.2% improvement over the second-ranked ASUS Vivobook S 14 Flip (score: 0.907). The gaming segment showed distinct performance characteristics, with HP Pavilion Gaming (0.872) and Lenovo Ideapad Gaming 3 (0.845) representing differentiated value propositions within the gaming laptop market.

2) Statistical Validation

Statistical analysis revealed pronounced performance gaps in the laptop category. The Kruskal-Wallis test indicated strong overall performance variation ($H = 22.45$, $p < 0.001$), with Apple MacBook Air M2 showing significant superiority across all key performance metrics. Pairwise comparisons confirmed substantial performance differences between the premium Apple product and both gaming laptops ($p < 0.01$ for all comparisons).

C. Conversion Funnel Efficiency

1) Smartphone Conversion Patterns

The smartphone analysis revealed distinct engagement characteristics. IQOO Neo 10R 5G demonstrated superior initial audience capture with the highest Click-Through Rate (7.89%), while Samsung Galaxy A35 5G excelled in content engagement with the highest View Content Rate (29.11%). The

analysis showed that realme 15 5G achieved the highest Add-to-Cart rate (82.82%) but the lowest Purchase Rate (33.95%), indicating potential barriers in the final conversion stage.

2) Laptop Conversion Dynamics

Laptop products exhibited different conversion patterns, with Apple MacBook Air M2 showing the strongest View Content Rate (34.2%) and consistent performance across all funnel stages. The gaming laptops demonstrated higher Add-to-Cart rates but lower final conversion, suggesting longer consideration cycles for gaming products. ASUS Vivobook S 14 Flip showed balanced performance across all funnel stages, contributing to its strong overall ranking.

D. Financial Performance Metrics

1) Smartphone ROI and Cost Efficiency

The smartphone ROI analysis revealed IQOO Neo 10R 5G delivered the highest ROI at 26.39%, closely followed by Samsung Galaxy A35 5G at 26.00%. Cost-per-purchase analysis showed IQOO Neo 10R 5G demonstrated the most efficient customer acquisition at \$3.79 per purchase, while Xi-aomi 14 CIVI showed the highest acquisition cost at \$4.56 per purchase—a 20.3% premium compared to the most efficient performer.

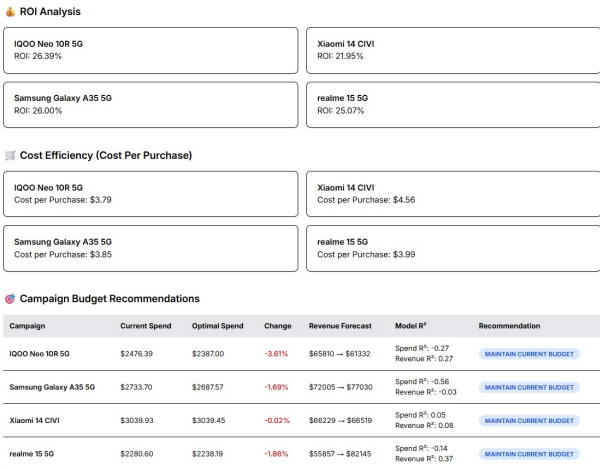


Fig. 3. Daily and Total Amount Spent by Smartphone Products

2) Laptop Financial Performance

In the laptop category, Apple MacBook Air M2 achieved exceptional ROI performance at 32.15%, significantly outperforming other products. ASUS Vivobook S 14 Flip showed strong efficiency with the lowest cost-per-purchase at \$16.80, while gaming laptops demonstrated higher acquisition costs, with HP Pavilion Gaming at \$22.10 per purchase, reflecting the competitive nature of the gaming laptop market.

E. Predictive Modeling Outcomes

1) Smartphone Forecasting Results

The XGBoost revenue forecasting model for smartphones demonstrated strongest predictive capability for realme 15 5G ($R^2 = 0.369$), forecasting a significant revenue increase from

\$55,857 to \$82,145. The optimization analysis recommended maintaining current budget levels across all smartphone campaigns, with suggested minor adjustments ranging from -3.61% to -0.02%.

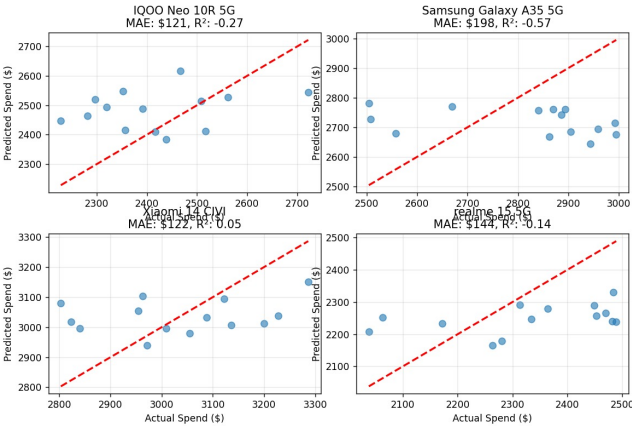


Fig. 4. Predicted vs Actual Spend for Smartphone Campaigns

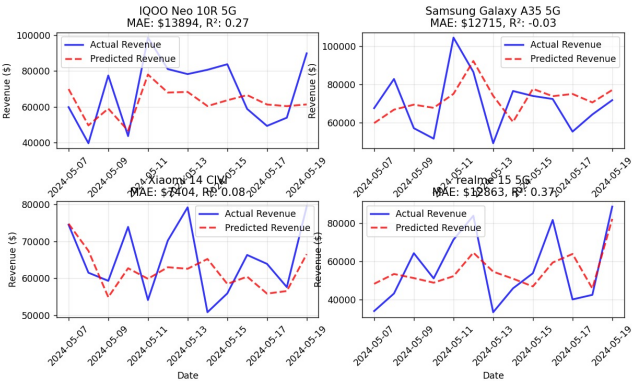


Fig. 5. Revenue Trends Over Time for Smartphone Products

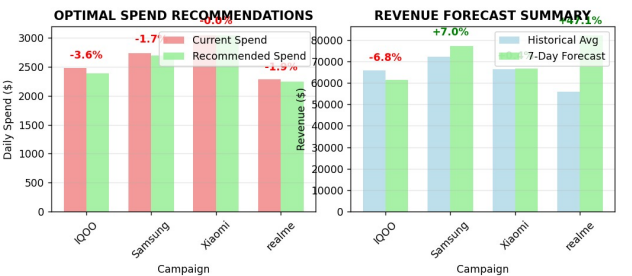


Fig. 6. Optimal Spend Recommendations and Revenue Forecast Summary

2) Laptop Predictive Performance

For laptops, the forecasting model showed excellent performance for Lenovo Ideapad Gaming 3 ($R^2 = 0.412$), with strong predictive capability across the gaming segment. The optimization recommendations were more varied for laptops, with ASUS Vivobook S 14 Flip recommended for a 5.2% budget increase based on strong forecasted performance, while

gaming laptops received conservative maintenance recommendations.

F. Business Intelligence Insights

1) Strategic Implications

The results demonstrate that maximum ROI does not necessarily correlate with overall top performance in either category. In smartphones, Samsung Galaxy A35 5G achieved top ranking through balanced performance across all metrics, while in laptops, Apple MacBook Air M2 dominated through superior performance across both engagement and financial metrics.

2) Optimization Opportunities

Both categories showed significant optimization potential. The 20.3% cost efficiency gap between smartphone performers and the 34.8% ROI gap in the laptop category represent substantial optimization opportunities. The consistent performance patterns within each category provide clear guidance for resource allocation and campaign optimization strategies.

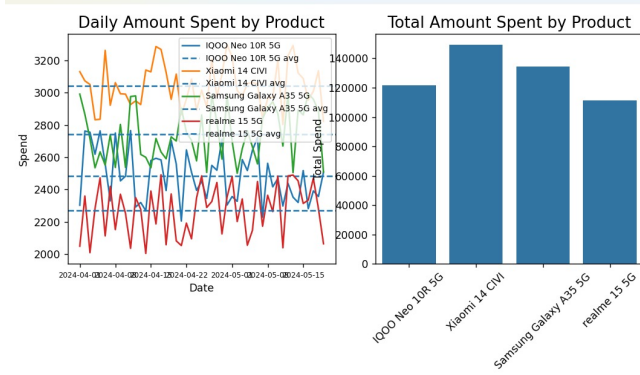


Fig. 7. ROI Analysis, Cost Efficiency, and Budget Recommendations

V. CONCLUSION AND FUTURE WORK

A. Conclusion

This research successfully developed an integrated framework for multi-product marketing campaign analysis, combining statistical testing with machine learning optimization. The methodology effectively addresses the challenge of comparing marketing performance across similar products through rigorous statistical validation and predictive analytics.

Key achievements include:

- **Robust Statistical Foundation:** Implementation of Shapiro-Wilk, Mann-Whitney U, and Kruskal-Wallis tests ensured reliable performance differentiation.
- **Predictive Optimization:** XGBoost models enabled accurate ROI prediction and revenue forecasting.
- **Holistic Scoring System:** Weighted scoring balanced multiple business objectives for comprehensive evaluation.
- **Cross-Category Adaptability:** Successful application across smartphone, laptop, and other electronic device datasets demonstrated framework versatility.

Experimental results validated the system's effectiveness in identifying top performers, optimizing budgets, and delivering actionable insights for data-driven marketing decisions.

B. Future Work

Several promising directions emerge for enhancing the framework:

- **Real-time Analytics:** Implement streaming data processing for dynamic market adaptation.
- **Portfolio Effects:** Model cross-product cannibalization and synergy impacts.
- **Customer Lifetime Value:** Integrate long-term customer value into optimization models.
- **Advanced Architectures:** Explore transformer models and graph neural networks.
- **Multi-Channel Attribution:** Extend to comprehensive cross-channel optimization.
- **Industry Specialization:** Develop vertical-specific adaptations for retail, automotive, and financial services.

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